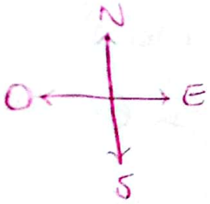
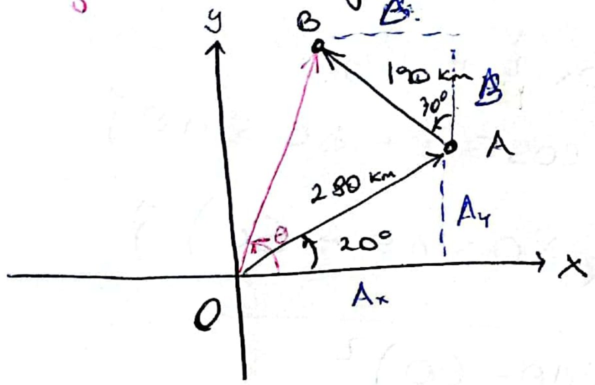
1— Avión $\Rightarrow$ 280 km OA, dirección 20° N-E
190 km AB, dirección 30° N-O



Gráfica



Vectores posición

$$\vec{A} = A_x \hat{\imath} + A_y \hat{\jmath}$$
$$= A \cos 20° \hat{\imath} + A \sin 20° \hat{\jmath}$$
$$= 280 \cos 20° \hat{\imath} + 280 \sin 20° \hat{\jmath}$$
$$= [263.1 \hat{\imath} + 95.76 \hat{\jmath}] \text{ Km}$$

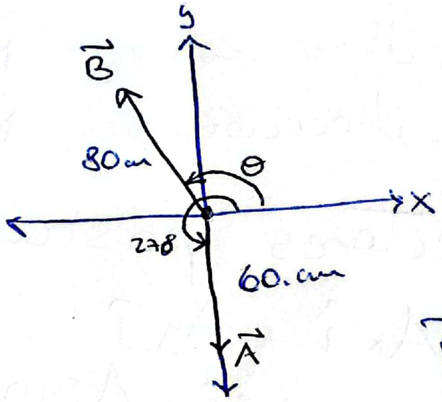
$$\vec{B} = B_x (-\hat{\imath}) + B_y \hat{\jmath} = - B \sin 30° \hat{\imath} + B \cos 30° \hat{\jmath}$$
$$= -190 \sin 30° \hat{\imath} + 190 \cos 30° \hat{\jmath}$$
$$= [-95 \hat{\imath} + 164.5 \hat{\jmath}] \text{ Km}$$

$$\vec{A} + \vec{B} = (A_x + B_x) \hat{\imath} + (A_y + B_y) \hat{\jmath} = \vec{C} = C_x \hat{\imath} + C_y \hat{\jmath}$$
$$= [168.1 \hat{\imath} + 260.3 \hat{\jmath}] \text{ Km}$$

$$|c| = \sqrt{C_x^2 + C_y^2} = \sqrt{(168.1)^2 + (260.3)^2} = \boxed{309.9 \text{ Km}}$$

$$\tan \theta = \frac{C_y}{C_x} \Rightarrow \theta = \tan^{-1}\left(\frac{C_y}{C_x}\right) = \tan^{-1}\left(\frac{260.3}{168.1}\right)$$
$$= \boxed{57°}$$

2 -



$\vec{A} = A_x\hat{i} + A_y\hat{j}$

$= A_y\hat{j} = A\sin 270°\hat{j}$

$= 60\sin 270°\hat{j} = -60\,cm$

$\vec{B} = B_x\hat{i} + B_y\hat{j}$

$= 80\cos\theta\,\hat{i} + 80\sin\theta\,\hat{j}$

$\Rightarrow \vec{A} + \vec{B} = 80\cos\theta\,\hat{i} + (80\sin\theta - 60)\hat{j}$

$\Rightarrow |\vec{A} + \vec{B}| = \sqrt{80^2\cos^2\theta + (80\sin\theta - 60)^2}$

$= \sqrt{80^2\cos^2\theta + 80^2\sin^2\theta - 2(80)(60)\sin\theta + 60^2}$ (a)

$= \sqrt{80^2 + 60^2 - 2(80)(60)\sin\theta} = \boxed{\sqrt{10000 - 9600\sin\theta}}$

(b) $\theta = 270°$ para que $|\vec{A} + \vec{B}|$ sea máximo, ya que $\sin 270° = -1$ y las componentes de la raíz quedan positivos

$\Rightarrow |\vec{A} + \vec{B}| = \sqrt{10000 + 9600} = \boxed{140\,cm}$

(c) $\theta = 90°$, para que $|\vec{A} + \vec{B}|$ sea mínimo, ya que $\sin 90 = +1$

$\Rightarrow |\vec{A} + \vec{B}| = \sqrt{10000 - 9600} = \boxed{20\,cm}$

$$3- \quad y = y_0 + \tan\theta(x - x_0) - \frac{1}{2} g \left(\frac{x - x_0}{v_0 \cos\theta}\right)^2$$

$$\text{Si } v_{0x} = \frac{\Delta x}{t} \Rightarrow t = \frac{x - x_0}{v_{0x}} = \frac{x - x_0}{v_0 \cos\theta}$$

$$\Delta y = v_{0y} t - \frac{1}{2} g t^2$$

$$\Longrightarrow y = y_0 + v_{0y}\left(\frac{x - x_0}{v_0 \cos\theta}\right) - \frac{1}{2} g \left(\frac{x - x_0}{v_0 \cos\theta}\right)^2$$

$$= y_0 + \cancel{v_0} \sin\theta \left(\frac{x - x_0}{\cancel{v_0} \cos\theta}\right) - \frac{1}{2} g \left(\frac{x - x_0}{v_0 \cos\theta}\right)^2$$

$$= \boxed{y_0 + \tan\theta\,(x - x_0) - \frac{1}{2} g \left(\frac{x - x_0}{v_0 \cos\theta}\right)^2}$$